

Every Quasiceutralizer Is Associative

A full answer to the first question in arXiv:math/0312501

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Status. `candidate_full_solution_likely_valid` for the first question in Remark 5.5.

Source questions. Kaneda–Paulsen ask whether every quasiceutralizer is associative and whether every associative quasiceutralizer belongs to $QMB(X)$. They conjecture that both answers are negative [1].

can be no completely isometric nonembeddings of (C_2, m_1) into an operator algebra. It is interesting to note that m_1 is also not a quasiceutralizer. In fact, it is not hard to show by a direct calculation that $QC(C_2) = QMB(C_2)$.

Remark 5.5. *Is every quasiceutralizer, automatically an associative bilinear map? Is every associative quasiceutralizer in $QMB(X)$? We conjecture that the answer is no to both of these questions, but we do not know of an example.*

Answer to the first question. Yes: every quasiceutralizer on every operator space is automatically associative. Thus the conjectured answer to the first question has the opposite sign. The second question is not needed for this conclusion and is not claimed solved here.

Definitions and canonical actions. A bilinear map $m : X \times X \rightarrow X$ is a quasiceutralizer if there are completely bounded maps

$$\gamma : X \longrightarrow M_l(X), \quad \psi : X \longrightarrow M_r(X)$$

such that

$$m(x, y) = \gamma(x)y = x\psi(y) \quad (x, y \in X). \quad (1)$$

In the injective-envelope realization used in the source, $X \subset I(X)$ is the 12 corner of the linking C^* -algebra $I(S_X)$, while $M_l(X) \subset I_{11}(X)$ and $M_r(X) \subset I_{22}(X)$. Consequently their canonical actions obey

$$(ax)b = a(xb) \quad (a \in M_l(X), x \in X, b \in M_r(X)), \quad (2)$$

simply by associativity in $I(S_X)$.

Proof intuition. Use the right-multiplier formula for the outer product in the left bracketing, and the left-multiplier formula for the outer product in the right bracketing. The two expressions differ only by parentheses in the linking algebra, so they coincide.

Theorem 1. *Let X be an operator space. Every quasiceutralizer $m : X \times X \rightarrow X$ is an associative bilinear map.*

Proof. Choose γ and ψ as in (1). For arbitrary $x, y, z \in X$, apply respectively the right and left versions of (1), followed by (2):

$$\begin{aligned} m(m(x, y), z) &= m(x, y)\psi(z) = (\gamma(x)y)\psi(z) \\ &= \gamma(x)(y\psi(z)) = \gamma(x)m(y, z) = m(x, m(y, z)). \end{aligned}$$

Thus m is associative. □

Internal cross-check. The same computation shows that γ is automatically the source’s left quasihomomorphism. Indeed, for every $z \in X$,

$$\gamma(x)\gamma(y)z = \gamma(x)(y\psi(z)) = (\gamma(x)y)\psi(z) = \gamma(m(x, y))z.$$

Faithfulness of the canonical multiplier action gives $\gamma(x)\gamma(y) = \gamma(m(x, y))$. Proposition 5.2 of the source then yields associativity independently; the right-sided calculation is identical.

The remaining second question. Since associativity is automatic, the second question becomes simply whether $QC(X) = QMB(X)$ for every operator space. Eight focused upgrade attempts did not settle it. Exact rational linear-algebra searches found equality for every connected matrix-unit support through 3×3 , all two-dimensional 0/1 subspaces of $M_{2,2}$, $M_{2,3}$, and $M_{3,2}$, and 400 deterministic random three-dimensional 0/1 subspaces of $M_{3,3}$. The general obstruction is that compatibility makes the relevant multiplier maps completely bounded, but does not visibly imply the special complete-contraction block estimates which the source uses to produce one fixed middle quasimultiplier.

Novelty and verification. Exact-sentence, title, arXiv, and core-term searches through 2026-08-13 found no later answer. The proof uses only the source’s definitions and canonical corner realization. Novelty confidence is moderate pending specialist review. The accompanying verification report checks every typing and corner multiplication and records the exact search scope.

Human-review recommendation. Verify that the standard left and right multiplier actions used in Definition 5.1 are the 11- and 22-corner actions in $I(S_X)$. Equation (2) then makes the proof immediate.

References

- [1] M. Kaneda and V. I. Paulsen, *Quasimultipliers of operator spaces*, J. Funct. Anal. 217 (2004), 347–365; arXiv:math/0312501.